FYUGP B.Sc. MATHEMATICS HONOURS

CONVEX OPTIMIZATION (MAT5EJ303 (2))

Semester V · Elective (Specialisation: Data Science)

MODULE 1: REVIEW OF MULTIVARIABLE CALCULUS

Model questions on topics the past papers miss

Section A: 1 mark each, answer all

Hessian matrix

1 Write the Hessian matrix of $f(x, y) = x^2y + 3y^2$. **(1 mark)**

Eigenvalues of the Hessian

2 State the condition on the eigenvalues of the Hessian at a stationary point which ensures that the point is a local minimum. **(1 mark)**

Total derivative and Jacobian matrix

3 Write the matrix that represents the total derivative $f'(\mathbf{a})$ of a differentiable map $f : \mathbb{R}^n \to \mathbb{R}^m$, and give its entries. **(1 mark)**

Section B: 2 marks each, answer all

Second order Taylor theorem

4 State the second order Taylor theorem for a real-valued function f of n variables. **(2 marks)**

Mean value theorem

5 State the mean value theorem for a differentiable real-valued function f on $\mathbb{R}^n$. **(2 marks)**

Section C: 4 marks each, answer all

Hessian matrix and symmetry

6 Let $f(x, y) = x^3 - 3xy + y^2$. Find all second order partial derivatives of f, and write the Hessian matrix at $(1, 1)$. State whether the Hessian is symmetric and give the reason. **(4 marks)**

Eigenvalues of the Hessian

7 Let $f(x, y) = x^2 + xy + y^2 - 3x$. Find the stationary point of f, the eigenvalues of the Hessian there, and classify the stationary point. **(4 marks)**

Section D: 10 marks, answer the question

Classification of stationary points

8 Find all stationary points of $f(x, y) = x^3 - 3x + 3y^2$, and classify each one using the Hessian matrix and its eigenvalues. **(10 marks)**

Key and mapping: Module I model paper

Section A: 1 mark each, answer all

1. (Unit 5). $H = \begin{pmatrix} 2y & 2x \\ 2x & 6 \end{pmatrix}$

2. (Units 7, 8). Both eigenvalues are positive (the Hessian is positive definite).

3. (Unit 2). The Jacobian matrix $f'(\mathbf{a}) = \left(\dfrac{\partial f_i}{\partial x_j}(\mathbf{a}) \right)$, with m rows and n columns.

Section B: 2 marks each, answer all

4. (Unit 6). If f is twice differentiable near $\boldsymbol{\xi}$, then for $\mathbf{x}$ near $\boldsymbol{\xi}$, $f(\mathbf{x}) = f(\boldsymbol{\xi}) + f'(\boldsymbol{\xi})(\mathbf{x} - \boldsymbol{\xi}) + \frac{1}{2}(\mathbf{x} - \boldsymbol{\xi})^T f''(\boldsymbol{\eta})(\mathbf{x} - \boldsymbol{\xi})$ for some $\boldsymbol{\eta}$ on the line segment joining $\mathbf{x}$ and $\boldsymbol{\xi}$.

5. (Unit 6). For $\mathbf{a}, \mathbf{b}$ with the segment between them in the domain, there is $\boldsymbol{\xi}$ on the segment with $f(\mathbf{b}) - f(\mathbf{a}) = f'(\boldsymbol{\xi})(\mathbf{b} - \mathbf{a})$.

Section C: 4 marks each, answer all

6. (Unit 5). $f_x = 3x^2 - 3y$, $f_y = -3x + 2y$; $f_{xx} = 6x$, $f_{xy} = f_{yx} = -3$, $f_{yy} = 2$. At $(1, 1)$: $H = \begin{pmatrix} 6 & -3 \\ -3 & 2 \end{pmatrix}$. It is symmetric because $f_{xy} = f_{yx}$ (equality of mixed partials, since f has continuous second order partials).

7. (Units 7, 8). $f_x = 2x + y - 3 = 0$ and $f_y = x + 2y = 0$ give $(x, y) = (2, -1)$. The Hessian is $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, with characteristic equation $(2 - \lambda)^2 - 1 = 0$, so $\lambda = 1$ and $\lambda = 3$. Both are positive, so $(2, -1)$ is a local minimum with value $f(2, -1) = -3$.

Section D: 10 marks, answer the question

8. (Units 4, 8). $f_x = 3x^2 - 3 = 0$ gives $x = \pm 1$; $f_y = 6y = 0$ gives $y = 0$. The stationary points are $(1, 0)$ and $(-1, 0)$. The Hessian is $\begin{pmatrix} 6x & 0 \\ 0 & 6 \end{pmatrix}$. At $(1, 0)$: $H = \operatorname{diag}(6, 6)$, eigenvalues $6, 6$, both positive; a local minimum with value $f(1, 0) = -2$. At $(-1, 0)$: $H = \operatorname{diag}(-6, 6)$, eigenvalues -6 and 6 of opposite sign; a saddle point with value $f(-1, 0) = 2$.

Mapping to the syllabus and Binmore (2nd ed.)

Question	Syllabus unit	Binmore section, printed pages
1	Unit 5 (Hessian matrix)	19.31 (second derivatives), printed p. 219
2	Units 7, 8 (eigenvalue test)	19.37–19.40, printed pp. 223–227
3	Unit 2 (total derivative, Jacobian matrix)	19.12–19.19, printed pp. 199–210
4	Unit 6 (second order Taylor)	19.36, printed p. 222
5	Unit 6 (mean value theorem)	19.35, printed p. 221
6	Unit 5 (Hessian matrix); Schwarz's theorem	19.31, printed p. 219; symmetry of mixed partials
7	Units 7, 8 (eigenvalues and classification)	19.37–19.40, printed pp. 223–227
8	Units 4, 8 (stationary points, classification)	19.27–19.30, printed pp. 217–218; 19.40, printed p. 227

Coverage: the Module I PYQ does not reach units 2 (total derivative), 5, 6, 7 and the classification part of 8. Questions 1 to 8 are drafted for those gaps; they are not taken from past papers.